

# Yasin Engin

Computer Engineering Student | Network Automation, Go Prototypes & DTN/LEO Experiments

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## EVIDENCE USED ON THIS SITE

Project pages distinguish available code and configuration from retained run output, checked results, study notes, and planned work. Limits are stated where a result has not yet been reproduced.

## PROFESSIONAL SUMMARY

Computer Engineering student at Ondokuz Mayıs University documenting network automation labs, Go service prototypes, and research-oriented experiments with Delay-Tolerant Networking and satellite-network emulation. Current work uses tools including Nokia SR Linux, Containerlab, Ansible, ION-DTN, NetSatBench, Go, gRPC, Docker, RabbitMQ, and PostgreSQL.

## TECHNICAL SKILLS

### Network & SDN

Cisco IOS

Nokia SR Linux

BGP

OSPF

VLAN / VRF

MPLS / VXLAN

### Network Automation

Containerlab

Ansible

YANG

gNMI

NETCONF / RESTCONF — studying

JSON-RPC

## Backend & Distributed Systems

Go

Python

REST APIs

gRPC / Protobuf

RabbitMQ

PostgreSQL / MongoDB

## DevOps & Observability

Linux

Docker / Swarm

Docker Compose

Caddy

Prometheus

Grafana

Git / GitHub

## Current Study Topics

NDN / ICN

LEO / NTN

DTN / ION-DTN

Network Orchestration

Distributed Control Plane

## FEATURED PROJECTS

### Network Automation Labs with Nokia SR Linux and Containerlab

Open Source

*Nokia SR Linux, Containerlab, Ansible, YANG/gNMI, JSON-RPC, Python*

- Built reproducible spine-leaf network automation labs with data-driven configuration and infrastructure-as-code habits.
- Connected SDN, network programmability, and observability concepts with hands-on lab validation workflows.

*Read case study*

### Network Inventory API Prototype

Open Source

*Go, PostgreSQL, JWT, RBAC, Docker, REST API*

- Built a learning-oriented inventory API for locations, devices, interfaces, VLANs, users, and audit records.
- Implemented authentication, role-based authorization, validation, filtering, pagination, and a local Docker setup.

*Read case study*

## **Tolerex**

**Open Source**

*Go, gRPC, mTLS, Leader-Based Replication, Prometheus, Grafana*

- Built a fixed-leader replicated-storage prototype with disk-backed persistence; leader election and leader failover are not implemented.
- Implemented mTLS gRPC communication, heartbeat-based member detection, crash-aware reads, and stress-test support.

*Read case study*

## **Netreka Nexus**

**Open Source**

*Go, RabbitMQ, Docker Swarm, Caddy, gRPC, PostgreSQL, MongoDB*

- Built a learning-oriented microservices study with broker and API-gateway boundaries.
- Implemented centralized logging, service decoupling, manual ACK/NACK consumer behavior, and polyglot persistence.

*Read case study*

## **NetSatBench Lab Guide**

**Future Networks**

*LEO Satellite Emulation, Docker, Etcd, VXLAN, Linux tc/netem, iperf3*

- Created a Turkish reproducible lab guide for NetSatBench-based LEO satellite network emulation.
- Documented worker configuration, overlay networking, routing tests, performance checks, and troubleshooting workflows.

*Read case study*

## **ION-DTN and NetSatBench Integration**

**Implementation + Selected Runs**

*ION-DTN, NetSatBench, BPv7, Contact Plans, OWLT, Linux tc/netem*

- Worked on ION node configuration, contact/range-plan generation, NetSatBench link scheduling, orbital geometry, OWLT, link-rate inputs, and guard-time calculations.

- Selected runs retain scripts or outputs; later constellation scenarios require separate revalidation.

*Read project notes*

## **Go Network Programming**

**Open Source**

*Go, TCP/UDP, HTTP, DNS, ICMP, CIDR, Sockets*

- Built protocol-level experiments around TCP/UDP clients and servers, HTTP services, diagnostics, and serialization.
- Practiced concurrency, timeouts, and instrumentation patterns for backend and network automation code.

*Read Go networking case study*

## **TECHNICAL TEACHING / COMMUNITY**

### **Founder / Technical Content Creator - Netreka Akademi**

**Public Teaching**

*CCNA, CCNP ENCOR, Network Automation, SDN, Nokia SR Linux, Containerlab*

- Publishes Turkish technical videos on network foundations, certification study, and laboratory workflows.
- Maintains a CCNP ENCOR series covering switching, routing, multicast, and troubleshooting topics.
- Documents introductory network-automation labs using Nokia SR Linux and Containerlab.

## **RESEARCH READING INTERESTS**

Maintains a public reading list covering network automation, YANG/gNMI, NDN/ICN, LEO/NTN systems, Delay-Tolerant Networking, network orchestration, and distributed systems. Listed topics are study areas, not claims of completed expertise.

## EDUCATION

### Computer Engineering

Current

Ondokuz Mayıs University, Turkey

## CERTIFICATIONS AND CURRENT STUDY

- Cisco Certified Network Associate (CCNA) — Cisco, April 2025
- CyberOps Associate course completion — Cisco Networking Academy, March 2024
- CCNP Enterprise study — in progress
- Nokia SR Linux Network Automation